STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 9: Mechanical Properties of Fluids

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Equivalent correct methods should receive credit.

Q1 1 mark

Normal force per unit area.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q2 1 mark

Pa s.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q3 1 mark

A1v1=A2v2.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q4 1 mark

p+rhov^2+rhogh=constant.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q5 1 mark

N/m.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q6 1 mark

rhogh.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q7 1 mark

6pietarv.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q8 1 mark

2S/r.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q9 2 marks

30,000 Pa.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q10 2 marks

Doubles.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q11 2 marks

Lower.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q12 2 marks

100 N.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q13 2 marks

40 Pa.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q14 2 marks

Bubble has two liquid-air surfaces; Deltap=4S/r.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q15 3 marks

8 m/s.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q16 3 marks

16,000 Pa.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q17 3 marks

~1.88 x 10- N.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q18 3 marks

Hydrostatic balance gives p=p+rhogh due to weight of fluid above.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q19 4 marks

4.5 m/s.

Suggested split: 1 for principle, 1 for equation, 1 for working and 1 for result/explanation. Accept an equivalent correct method.

Q20 4 marks

Deltap=4S/r=60 Pa.

Suggested split: 1 for principle, 1 for equation, 1 for working and 1 for result/explanation. Accept an equivalent correct method.